

FULL VELOCITY ESTIMATION DOCUMENT FOR TWIN T100 THRUSTER
CATAMARAN

System Overview

Parameter	Value	Source
Thruster Model	2 × T100	Blue Robotics
Operating Voltage	12.8 V	Estimated during power requirement analysis
Max Combined Current	23 A	Estimated during power requirement analysis
Max Power (Thrusters only)	294.4 W	Estimated during power requirement analysis
System Power with Margin	408.4 W	Estimated during power requirement analysis (+25% margin)
Water Density (ρ)	1000 kg/m³	Fluid Dynamics Standard
Drag Coefficient (C _D)	0.6	Assumed for catamaran-style hull
Projected Frontal Area (A)	0.05 m²	Estimated based on 1.5 m hull, 25 cm draft

Manufacturer-Based Thrust and Power Data

PWM (µs)	Power (W, 2 Thrusters)	Thrust (N, 2 Thrusters)	Source
1550	30	2.0	Blue Robotics T100 Docs
1600	60	4.2	Blue Robotics
1650	120	7.0	Blue Robotics
1700	180	10.0	Blue Robotics
1750	240	13.0	Blue Robotics
1800	294	15.6	Blue Robotics
1850	~350	17.4	Blue Robotics
1900	~400	19.0	Blue Robotics

Velocity Calculation (Ideal Conditions)

Drag-Based Equation:

$$v = \sqrt{\frac{2T}{\rho \cdot C_D \cdot A}} = \sqrt{0.667 \cdot T}$$

Velocity Calculations (Ideal and Realistic)

PWM (µs)	Thrust (N)	Ideal Velocity (m/s)	Adjusted Velocity (−25%)	Power (W)
1550	2.0	$\sqrt{0.1344} = \mathbf{0.365}$	0.274 m/s	30
1600	4.2	$\sqrt{0.2801} = \mathbf{0.529}$	0.397 m/s	60
1650	7.0	$\sqrt{0.4669} = \mathbf{0.683}$	0.512 m/s	120
1700	10.0	$\sqrt{0.667} = \mathbf{0.817}$	0.613 m/s	180

1750	13.0	$\sqrt{0.8671} = \mathbf{0.931}$	0.698 m/s	240
1800	15.6	$\sqrt{1.0405} = \mathbf{1.020}$	0.765 m/s	294
1850	17.4	$\sqrt{1.1616} = \mathbf{1.078}$	0.808 m/s	~350
1900	19.0	$\sqrt{1.2673} = \mathbf{1.125}$	0.844 m/s	~400